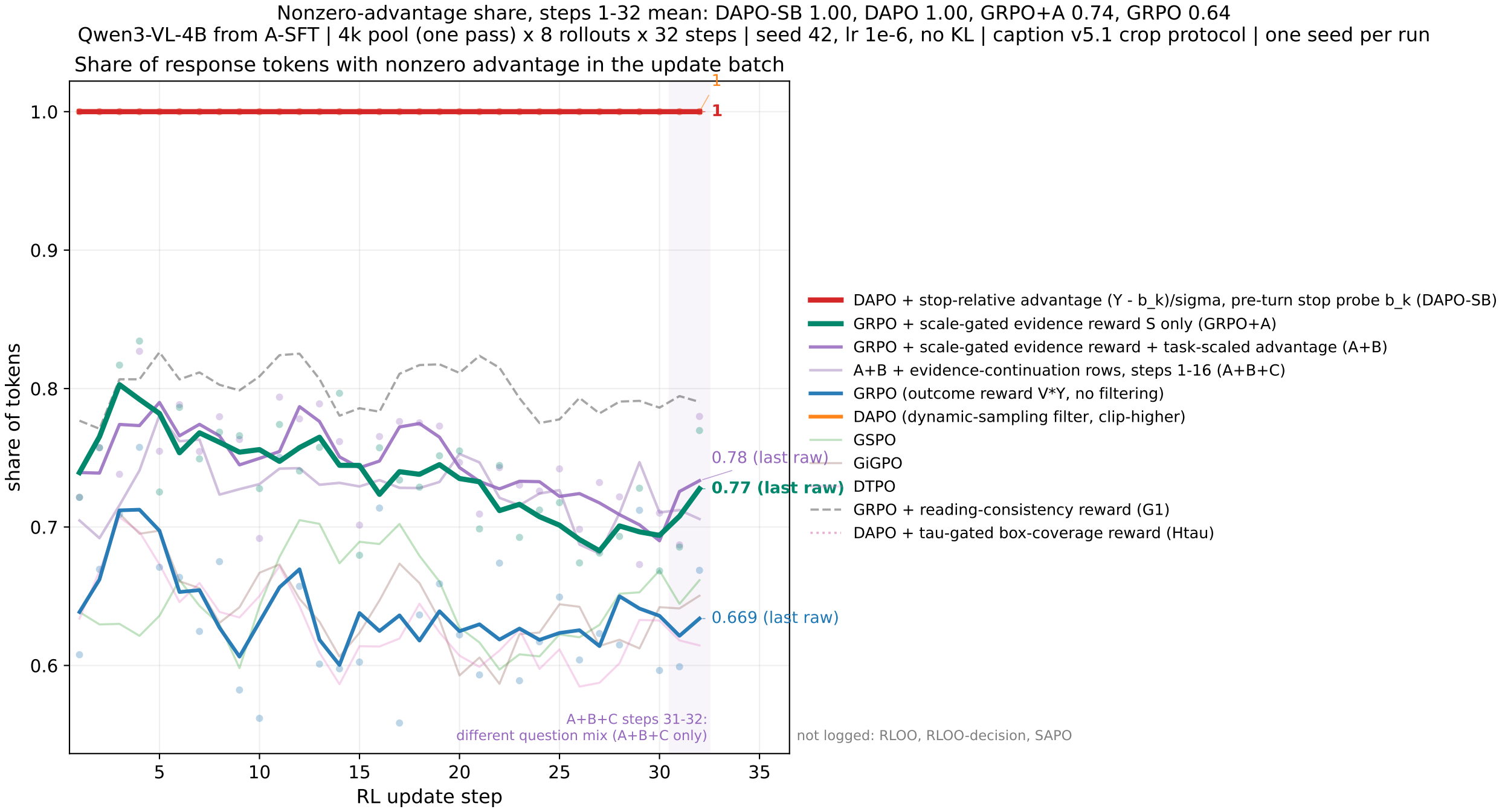
Nonzero-advantage share, steps 1-32 mean: DAPO-SB 1.00, DAPO 1.00, GRPO+A 0.74, GRPO 0.64
Qwen3-VL-4B from A-SFT | 4k pool (one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, no KL | caption v5.1 crop protocol | one seed per run
Share of response tokens with nonzero advantage in the update batch
share of tokens
RL update step
1.0
0.9
0.8
0.7
0.6
5
10
15
20
25
30
35
1
1
0.78 (last raw)
0.77 (last raw)
0.669 (last raw)
A+B+C steps 31-32:
different question mix (A+B+C only)
DAPO + stop-relative advantage (Y - b_k)/sigma, pre-turn stop probe b_k (DAPO-SB)
GRPO + scale-gated evidence reward S only (GRPO+A)
GRPO + scale-gated evidence reward + task-scaled advantage (A+B)
A+B + evidence-continuation rows, steps 1-16 (A+B+C)
GRPO (outcome reward V*Y, no filtering)
DAPO (dynamic-sampling filter, clip-higher)
GSPO
GiGPO
DTPO
GRPO + reading-consistency reward (G1)
DAPO + tau-gated box-coverage reward (Htau)
not logged: RLOO, RLOO-decision, SAPO